

MIN JONG LEE

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Public curriculum vitae | Record updated September 27, 2026

RESEARCH IDENTITY

Interface and device physics for emerging electronics

Research interests: molecular and thin-film interfaces, defects, ionic dynamics, charge transport, emerging memory, optoelectronics, and flexible-device R&D. Integrated electronic systems are a long-term research direction.

EDUCATION

Integrated M.S.-Ph.D. Program in Electrical Engineering | March 2023 - Present

Korea University, Seoul, Republic of Korea

GPA: 4.06 / 4.50

Advisor: Prof. Jae Won Shim

B.S. in Electronic Engineering | March 2019 - February 2023

Soongsil University, Seoul, Republic of Korea

GPA: 4.01 / 4.50 | Magna Cum Laude

FIRST-AUTHOR PUBLICATIONS

1. Min Jong Lee, Sang Heon Lee, Dong Gyu Lee, Tae Hyuk Kim, Yubhin Cho, Gyeong Min Lee, Sung Su Yoon, Seon Joong Kim, Hyungju Ahn, Tae Kyung Lee*, and Jae Won Shim*. "Linear and Symmetric Artificial Synapses Driven by Hydrogen Bonding for Accurate and Reliable Neuromorphic Computing." **Advanced Materials** · **37(45)**, e11728 (2025). doi:10.1002/adma.202511728

2. Min Jong Lee, Tae Hyuk Kim, Sang Heon Lee, Seunghyun Oh, Muhammad Asghar Khan, Gyeong Min Lee, Young Kyun Choi, Soyeon Lee, Hyungju Ahn, Soong Ju Oh, Jiwoong Yang, and Jae Won Shim*. "Intermediate Layer-Assisted Trap Density Reduction in Low-Power Optoelectronic Memristors for Multifunctional Systems." **Advanced Functional Materials** · **35(22)**, 2421080 (2025). doi:10.1002/adfm.202421080

3. Min Jong Lee, Seunghyun Oh, Muhammad Ahsan Saeed, Yelim Kang, Gyeong Min Lee, Hyungju Ahn, Yunsang Kim, and Jae Won Shim*. "Fine-tuning Hole Collection via Metal Pseudo-halogenated Thin-film Treatment for Propelling Advancements in Versatile platforms." **IEEE Transactions on Electron Devices** · **71(8)**, 4723-4731 (2024).

4. Min Jong Lee, Ji-Sang Park, Tae Hyuk Kim, Muhammad Ahsan Saeed, Gyeong Min Lee, and Jae Won Shim*. "Tailoring Hole-Selective Contacts via Self-Assembled Monolayers for advancing Indoor Organic Photovoltaic and Capacitor Devices." **Chemical Engineering Journal** · **481**, 148481 (2024). doi:10.1016/j.cej.2023.148481

5. **Min Jong Lee**, Muhammad Ahsan Saeed, Tae Hyuk Kim, Jae Jin Chung, and Jae Won Shim*. "Atomic layer deposition-processed laminated In₂O₃:V₂O₅ nanocompounds: Versatile electrode platforms for high-performance organic photovoltaics and capacitor device." **Journal of Alloys and Compounds** · **968**, 171951 (2023). doi:10.1016/j.jallcom.2023.171951

CO-AUTHORED PUBLICATIONS

6. Seon Joong Kim†, Junhyeok Park†, Byung Ku Jung, Ohhyun Kwon, Sang Heon Lee, Hyung Jin Choi, Seongyong Hong, **Min Jong Lee**, Soong Ju Oh*, and Jae Won Shim*. "Defect-Healed Perovskite Quantum Dots for Optoelectronic Applications in Indoor Photovoltaics and Photoplethysmography." **Small · Early View** (2026). doi:10.1002/sml.74660

7. Sung Su Yoon, Sang Heon Lee, **Min Jong Lee**, Seon Joong Kim, Hyungju Ahn, Han Jung Park and Jae Won Shim*. "Ionic Landscape Engineering via Perovskite Quantum Dots for Reliable and Energy-Efficient Perovskite Memristors." **ACS Nano** · **20(21)**, 15336-15346 (2026).

8. Ohhyun Kwon†, Jae Hoon Sont†, Tae Hyuk Kim, Gyeong Min Lee, Seunghyun Oh, Gayoung Ham, **Min Jong Lee**, Sang Heon Lee, Seon Joong Kim, Hyojung Cha, Han Young Woo*, and Jae Won Shim*. "Suppression of Photo-Mediated Traps in Integrated Organic Photovoltaic-Photodetector Devices via N-Type Self-Assembly-Driven Interfacial Engineering." **Advanced Functional Materials** · **36(28)**, e24759 (2026).

9. Minju Yang, Seon Joong Kim, Tae Hyuk Kim, Hyungju Ahn, **Min Jong Lee**, Yunsang Kim, and Jae Won Shim*. "Bi-Synergistic Ligand-Mediated Passivation of Surface Defects for Highly Efficient and Stable Cesium-Lead-Iodide Perovskite Quantum Dot Photovoltaics." **Materials Horizons** · **13**, 1906-1917 (2026).

10. Hee Chun Kim,† Ji Hyeon Lee,† Tae Hyuk Kim, **Min Jong Lee**, Eun Soo Shim, Seunghwan Bae, Hyungju Ahn, Han Jung Park, Nak-Seung Patrick Hyun, Jea Woong Jo,* and Jae Won Shim*. "Molecular Spacer Engineering of Carbazole-Phosphonic Acid SAMs: Probing Interfacial Energetics in Low-Light Organic Optoelectronic Devices." **Chemical Engineering Journal** · **527**, 171902 (2026). doi:10.1016/j.cej.2025.171902

11. Muhammad Asghar Khan, Yelim Kang, Muhammad Farooq Khan, Shania Rehman, **Min Jong Lee**, Sang Heon Lee, Seunghyun Oh, Tae Hyuk Kim, Seon Joong Kim, Hyungju Ahn, and Jae Won Shim*. "Ultraviolet light-Driven Artificial Neuromorphic Properties in Organic Transistors for Reservoir Computing and Encrypted Communication." **IEEE Transactions on Electron Devices** · **72(11)**, 6341-6348 (2025).

12. Sang Heon Lee, **Min Jong Lee**, Hyungju Ahn, Ji-Sang Park, Seon Joong Kim, Kihyun Lee, Tae Hyuk Kim, Gyeong Min Lee, Kwanpyo Kim, and Jae Won Shim*. "Synaptic Plasticity Modulation in Dion-Jacobson Perovskite Artificial Synapses Enabled by Crystallographic Orientation Control." **Small** · **21(34)**, 2504328 (2025).

13. Yelim Kang, Tae Hyuk Kim, **Min Jong Lee**, and Jae Won Shim*. "Dual-Mode Organic Photodetectors with Optimized Charge Dynamics and Barrier Modulation for Superior Spectral Selectivity." **Small** · **21(29)**, 2503701 (2025).

14. Tae Hyuk Kim,† Sang Young Jeong,† Seunghyun Oh, Yelim Kang, **Min Jong Lee**, Min Hun Jee, Han Young Woo,* and Jae Won Shim*. "Control of chemical doping-mediated space charge for energetic band-switching modulation in low-noise shortwave-infrared organic photodetector." **Advanced Materials** · **37(30)**, 2500126 (2025).

15. Seon Joong Kim, Muhammad Asghar Khan, Dong Gyu Lee, Tae Hyuk Kim, **Min Jong Lee**, Yelim Kang, Minju Yang, Hyungju Ahn, Kyoungwon Park, Tae Kyung Lee*, and Jae Won Shim*. "Strategic Selection of Functional Groups for Ligand Passivation to Improve Performance of Cesium-Lead-Iodide Quantum Dot Photovoltaics and Luminescent Solar Concentrators." **Advanced Functional Materials** · **35(33)**, 2423796 (2025).

16. Gyeong Min Lee, Tae Hyuk Kim, Gayoung Ham, **Min Jong Lee**, Seunghyun Oh, Yelim Kang, Hyungju Ahn, Hyojung Cha, Seungjun Chung, and Jae Won Shim*. "Enhancing Dynamic Range in Low-Noise 2D-Integrated Organic Photodiodes by Mitigating Langevin Recombination." **ACS Nano** · **19(4)**, 4650-4662 (2025).
17. Muhammad Ahsan Saeed†, Zubair Khalid†, **Min Jong Lee**, Hyungju Ahn, Jae-Min Oh*, and Jae Won Shim*. "Bidentate mode coordinated Ti3CNTx MXene-controlled interfacial engineering enables high-performance indoor organic photovoltaic devices." **Chemical Engineering Journal** · **495**, 153404 (2024).
18. Tae Hyuk Kim†, Ji Hyeon Lee†, Min Ho Jang†, Gyeong Min Lee, Eun Soo Shim, Seunghyun Oh, Muhammad Ahsan Saeed, **Min Jong Lee**, Byoung-Soo Yu, Do Kyung Hwang, Chae Won Park, Sae Youn Lee*, Jea Woong Jo*, and Jae Won Shim*. "Atto-scale noise near-infrared organic photodetectors enabled by controlling interfacial energetic offset through enhanced anchoring ability." **Advanced Materials** · **36(40)**, 2403647 (2024). doi:10.1002/adma.202403647
19. Seunghyun Oh†, Ye Lim Kang†, Tae Hyuk Kim, Seon Joong Kim, **Min Jong Lee**, Gyeong Min Lee, Muhammad Ahsan Saeed*, and Jae Won Shim*. "Enhancing the indoor performance of organic photovoltaic devices: Interface engineering with an aminobenzoic-acid-based self-assembled monolayer." **Journal of Physics: Energy** · **6**, 025015 (2024).
20. Tae Hyuk Kim†, Young Kyun Choi†, Gyeong Min Lee, Muhammad Ahsan Saeed, Byung Ku Jung, **Min Jong Lee**, Hyung Jin Choi, Soong Ju Oh*, and Jae Won Shim*. "Ultra-low Noise Level Infrared Quantum Dot Photodiodes with Self-Screenable Polymeric Optical Window." **Advanced Materials** · **36(4)**, 2309028 (2024).

MANUSCRIPT IN REVISION

Min Jong Lee, Hyoungwook Cho, Hyung-Min Lee*, and Jae Won Shim*, et al.. "Chiral Neuromorphic Memory IC."
Manuscript in revision | 2026; not published.

PATENTS

1. Ion-shielding films for inorganic light-emitting displays

금속 유사할로겐화 박막을 이용한 무기 발광 디스플레이용 이온 차폐막 및 그 제조방법

심재원, 이민종, 최학영, 장성환

Registered | Korea | Registration number: 10-2995909 | 2026.07.22

2. Multifunctional low-power optoelectronic memristors

다기능 저전력 광전자 멤리스터 장치 및 이의 제조 방법

심재원, 이민종

Registered | Korea | Registration number: 10-2968668 | 2026.05.22

3. Multifunctional low-power optoelectronic memristors

MULTIFUNCTIONAL LOW-POWER OPTOELECTRONIC MEMRISTORS DEVICE AND METHOD OF MANUFACTURING THEREOF

심재원, 이민종

Application | US | Application number: 19/549,394 | 2026.02.25

4. Semiconductor device

Semiconductor device

김태균, 심재원, 이민중

Application | US | Application number: 19/002282 | 2025.11.13

China and Taiwan are also listed; the identifier shown is for the US application.

5. Method of manufacturing semiconductor memory devices

METHOD OF MANUFACTURING SEMICONDUCTOR MEMORY DEVICE

김태균, 심재원, 이민중

Application | US | Application number: 19/037,592 | 2025.01.27

Taiwan is also listed; the identifier shown is for the US application.

6. Perovskite-polymer artificial synapses

페로브스카이트-고분자 박막을 이용한 선형 대칭형 인공 시냅스 소자 및 이의 제조 방법

심재원, 이민중

Application | Korea | Application number: 10-2025-0091536 | 2025.07.08

7. Method of manufacturing semiconductor memory devices

반도체 메모리 소자의 제조 방법

김태균, 심재원, 이민중

Application | Korea · US · Taiwan | Korean application number: 10-2024-0088513 | 2024.07.04

8. MIM capacitor fabrication using high-work-function $\text{In}_2\text{O}_3/\text{V}_2\text{O}$ nanolaminate electrodes to reduce leakage current

누설전류를 줄이기 위하여 인듐 산화물 및 바나듐 산화물의 나노라미네이트 고 일함수 전극을 통한 MIM 커패시터 제작 방법

김태균, 심재원, 이민중

Application | Korea · US · China · Taiwan | Korean application number: 10-2024-0060762 | 2024.05.08

RESEARCH PROJECTS

Multilevel precision limits in ion-migration-controlled memristors

상태 형성 동역학 및 저주파 잡음 해석 기반 이온 이동 제어형 멤리스터의 멀티 레벨 정밀도 한계 규명

Doctoral Research Support Program · Ministry of Education

Program period: 2026.09-present | Role: Principal Investigator

Photonic skin for ultralight aerial Physical AI

초경량 비행형 Physical AI를 위한 Photonic Skin 연구실

Basic Research Laboratory · MSIT

Program period: 2026.07-present | Role: Participating Researcher

Interconnect materials for highly flexible FPCBs

초고굴곡 FPCB 배선 소재 개발

Samsung Electronics

Program period: 2026.01-present | Role: Participating Researcher

Low-power infrared gas-sensing technology for smart homes

스마트 주거 환경을 위한 저전력 다중 복합 유해가스 측정용 적외선 가스센서 기술 개발

K-Sensor R&D · Ministry of Trade, Industry and Energy

Program period: 2024.04-present | Role: Participating Researcher

Overcoming low-light power-generation limits in photovoltaics

전일 동작 광전지 개발을 위한 약광 발전 한계 돌파 연구

Mid-career Researcher Program · MSIT

Program period: 2022.03-present | Role: Participating Researcher

ALD-based ultrathin multicomponent electrodes as alternatives to TiN

TiN 전극 대체를 위한 ALD 기반 초박막 다성분계 전극 개발

Samsung Electronics

Program period: 2020.09-2025.09 | Role: Participating Researcher

Training for the transition to inorganic light-emitting displays

산업 전환형 무기발광 디스플레이 전문인력양성사업

Ministry of Trade, Industry and Energy

Program period: 2023.03-2025.08 | Role: Participating Researcher

Durable stretchable adhesive films and large-area coating processes

고신축특성을 가지는 고신뢰성 고내구성 점착필름 및 대면적 코팅 공정 기술 개발

Ministry of Trade, Industry and Energy

Program period: 2020.04-2024.12 | Role: Participating Researcher

Stretchable PCBs using liquid metal particles

Liquid Metal Particle활용 Stretchable PCB 제작

Samsung Electronics

Program period: 2024.08-2024.11 | Role: Participating Researcher

AWARDS & ACADEMIC PROGRAMS

2026 | Outstanding Graduate Research Achievement, College of Engineering, Korea University

고려대 공과대학 연구실적 우수대학원생 수상

2026 | BK21 Recognition for Publication in an Outstanding Journal

BK21 참여대학원생 우수저널 게재 논문 선정

2026 | KU Hope Scholarship

KU희망장학금 선정

2025 | Excellence Award, Samsung Electronics Industry-Academic Cooperation Paper Award

삼성전자 산학협력 우수논문 우수상

2025 | Excellence Award for Next-Generation Engineering Researchers, IPESK

한국이공학진흥원장 차세대 공학연구자 우수상

2025 | Selected for the 2nd IPESK Next-Generation Engineering Researchers Program

한국이공학진흥원장 제2기 IPESK 차세대 공학 연구자 선정

Public version updated September 2026. Project dates are official program periods, not individual participation dates. English labels for Korean titles are descriptive translations. Author symbols: † equal contribution; * corresponding author. Phone number and detailed postal address omitted.